

widthTopological Memory.

Introduction Today:

Proof.

Walkthrough the math

Our Code

Code In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 b

Local Decoder

definition We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.

There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority

Storing Logical State - Classical Statement

Logical State For a random variable X we denote by:

$\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p -depolarizing channel.

$\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p -depolarized channel followed by an appl

The Theorem - Classical Statement Theorem There is $p_{th} \in (0, 1)$ such that for any $p < p_{th}$ and any bounded funct

Lemma There exists a constant $c > 0$, such: $\mathbf{E}_{\sim 2p(1-p)}[f \circ D] \leq \mathbf{E}_{\sim p}[f] + |f|e^{-c\sqrt{n}}$

Theorem Proof By induction on t . $\mathbf{E}_{\sim q^{(t)}}[f] = \mathbf{E}_{\sim q^{(t-1)}}[\mathbf{E}_{\sim p}[f \circ D]]$

$$\leq \mathbf{E}_{\sim p}[\mathbf{E}_{\sim p}[f \circ D]] + (t-1)|\mathbf{E}_p[f \circ D]|e^{-c\sqrt{n}}$$

$$\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}}$$

$$\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}} \text{ When in the last transition we used the lemma.}$$

Dependence Type I [h] [scale=0.7]

node (u1) at (3,3) [draw,shape=rectangle]u₁; node (x) at (0,6) [draw,shape=circle]x; node (y) at (0,0) [draw,shape=

$\Pr[(x, y) \in e|u_1 \text{ is satisfied}]$

$$= \Pr[x \in e|u_1 \text{ is satisfied}] \Pr[y \in e|u_1 \text{ is satisfied}]$$

$$\leq \Pr[x \in e] \Pr[y \in e]$$

Dependence Type II [h] [scale=0.7]

node (u1) at (3,1) [draw,shape=rectangle]u₁; node(u2) at (3,4) [draw,shape=rectangle]u₂; node (x) at (0,6) [draw,s

$\Pr[(x, y) \in e|z \text{ is not flipped}]$

$$= \Pr[x \in e|z \text{ is not flipped}] \Pr[y \in e|z \text{ is not flipped}]$$

$$\leq \Pr[x \in e] \Pr[y \in e]$$

widthheightblockDefinition Consider the moment after the application of the depolarizing channel, and right before

For a subset of qubits S , we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

Either x and y share a check that is not satisfied. (Type I)

Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)

Dependence

Claim Consider the application of a $2p(1-p)$ -depolarizing channel. Let S be a subset of qubits chosen uniformly ra

Dependence

Proof First, for an arbitrary S , denote by J the qubits that depend on qubits in S . Then, the expected number of

$$\leq \mathbf{E}_{\sim 2p(1-p)}[|J|] \leq 2p(1-p)\Delta^l|S|$$

Similarly, let J' be the qubits in S that connected to unsatisfied check, then:

$$\mathbf{E}_{\sim 2p(1-p)}[Type I \text{ dependent qubits in } S]$$

$$\leq \mathbf{E}_{\sim 2p(1-p)}[|J'|] \leq \frac{1-(1-4p(1-p))^{\Delta^2}}{2} \Delta^l|S|$$

Dependence We are going to use the following inequality: [width=]ineq.png

Dependence Thus, the probability that the number of dependent qubits in S is $|S|^{\frac{3}{4}}$ far from the expectation: $\Pr$

$$\text{So, for any } |S| > \alpha n \text{ we have: } \Pr_{\sim 2p(1-p)}[\# - \mathbf{E}[\#] \geq |S|^{\frac{3}{4}}] \leq e^{-\frac{2\alpha^{\frac{3}{4}}}{\Delta^l} \sqrt{n}}$$

Dependence Thus for small enough $p \in (0, 1)$, with probability greater than $1 - 2e^{-\frac{2\alpha^{\frac{3}{4}}}{\Delta^l} \sqrt{n}}$, the number of dependen

Lemma Proof Denote by S the subset of the flipped qubits, and denote by $de(S)$ the number of dependent qubits in

$$\leq \sum_{de(S) < \frac{1}{100}|S||S| \geq \alpha n} f \circ D(S) \cdot \Pr[S] + 2|f|e^{-c\sqrt{n}}$$

$$\leq \sum_{de(S) < \frac{1}{100}|S||S| \geq \alpha n} f(S)M(2p(1-p))^{\frac{99}{100}|S|} + 2|f|e^{-c\sqrt{n}}$$

$$\text{Lemma Proof } \leq \sum_{de(S) < \frac{1}{100}|S||S| \geq \alpha n} f(S)p^{|S|} + 2|f|e^{-c\sqrt{n}}$$

$$\leq \sum_S f(S)p^{|S|} + 2|f|e^{-c\sqrt{n}}$$

$$\leq \mathbf{E}_{\sim p}[f(S)] + 2|f|e^{-c\sqrt{n}}$$

Storing Logical State - Quantum Statement

Logical State Let $\rho^{(0)}$ be a mixed state over codewords. Define:

$$\rho^{(t)} = D \circ \mathcal{N}_p \rho^{(t-1)}$$

$\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p -depolarizing channel.

$\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p -depolarized channel followed by an appl

The Theorem - Quantum Statement Theorem There is $p_{th} \in (0, 1)$ such that for any $p < p_{th}$ and any bounded Herm

Lemma There exists a constant $c > 0$, such: $\mathbf{Tr} X D \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) \leq \mathbf{Tr} X \mathcal{N}_p(\rho) + 2\|X\|e^{-\Theta(\sqrt{n})}$

Theorem Proof By induction on t . $\mathbf{Tr} X \rho^{(t)} = \mathbf{Tr} X D \circ \mathcal{N}_p \rho^{(t-1)}$

$$\leq \mathbf{Tr} X D \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) + (t-1)\|X\|e^{-\Theta(\sqrt{n})}$$

$$\leq \mathbf{Tr} X \mathcal{N}_p(\rho) + t\|X\|e^{-\Theta(\sqrt{n})} \text{ When in the last transition we used the lemma. Corollary}$$

$$\mathbf{Tr} (I - X) \rho^{(t)} \leq \mathbf{Tr} (I - X) \mathcal{N}_p(\rho) + t\|I - X\|e^{-\Theta(\sqrt{n})}$$

$$\Leftrightarrow \mathbf{Tr} X \rho^{(t)} \geq \mathbf{Tr} X \mathcal{N}_p(\rho) - t\|I - X\|e^{-\Theta(\sqrt{n})}$$

Corollary II